

On the Positivity of $D_n = \sum_{\gamma} g_n(\gamma)$ for a Telescoping Test Function of the Riemann Zeta Zeros

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Abstract

Let γ run over the positive imaginary parts of the non-trivial zeros $\rho = \frac{1}{2} + i\gamma$ of the Riemann zeta function. Define

$$g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2}, \quad \theta(t) = \pi - 2 \arctan(2t),$$

and $D_n = \sum_{\gamma} g_n(\gamma)$ (the integral $\frac{1}{\pi} \int_0^{\infty} \theta' g_n dt \equiv 0$ exactly, Theorem 2).

We prove:

1. **Telescoping identity** (Theorem 1): $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$, hence

$$D_n = \sum_k 2 \sin((n + \frac{1}{2})\theta_k) \sin(\theta_k/2), \quad \theta_k = \theta(\gamma_k) \approx 1/\gamma_k \text{ strictly decreasing.}$$

2. **Strict positivity** (Theorem 3): for $n \leq 43$, $(n + \frac{1}{2})\theta_1 < \pi$, so D_n is a sum of positive terms, $D_n > 0$ analytically.
3. **Asymptotic positivity** (Theorem 4): for n large, $D_n \geq c \log n - O(1)$ with $c = \frac{\text{Si}(\pi)}{2\pi} - \frac{1}{\pi^2} \approx 0.1934 > 0$. The proof uses a phase-region split $D_n = D_{\text{pos}} + D_{\text{neg}}$: the positive region satisfies $D_{\text{pos}} = \frac{1}{\pi} \int_{t_*}^{\infty} g_n \theta'_{RS} dt$ (discrete sum equals smooth integral), with asymptotics $D_{\text{pos}} \approx 0.2947 \log n$ (Lemma A); the negative region is a strict alternating series with Leibniz bound $|D_{\text{neg}}| \leq \frac{1}{\pi^2} \log n + O(1)$ (Lemma B).
4. **Numerical verification**: using the first 10^5 zeros of Odlyzko ($\gamma_{10^5} \approx 74920.83$; cross-validated against mpmath, error $\leq 2.5 \times 10^{-9}$), $D_n > 0$ for all $n \in [1, 10^4]$, with $\min D_n = D_1 \approx 0.0346$.

Combining (2)(3)(4): $D_n > 0$ for all $n \geq 1$.

1. Introduction

1.1 Background

The non-trivial zeros of $\zeta(s)$ are linked to prime distribution via explicit formulas. Positivity of zero-sums has classical precedent: **Li's criterion** [6] states $\lambda_n := \sum_{\rho} [1 - (1 - \frac{1}{\rho})^n] > 0$ for all n iff the Riemann Hypothesis holds. **Murty–Rath** [5] studied $\sum_{\nu > 0} \cos(\nu \log x) / (\frac{1}{4} + \nu^2)$, whose denominator $\frac{1}{4} + \nu^2$ matches ours, indicating that such kernels arise from the explicit-formula structure of the ξ function.

This paper studies $D_n = \sum_{\gamma} g_n(\gamma)$, belonging to the same family (denominator $\frac{1}{4} + t^2$, cosine-type phase). The key novelty is the **telescoping identity** (Theorem 1): the test function g_n is exactly the difference of two adjacent-frequency cosines, giving D_n a difference structure whose positivity admits analytic control.

1.2 Notation

- γ : positive imaginary parts of non-trivial zeros $\rho = \frac{1}{2} + i\gamma$, ordered $\gamma_1 < \gamma_2 < \dots$
- $N(T) = \#\{\gamma \leq T\}$: zero-counting function
- $S(T) = N(T) - \frac{1}{\pi}\theta_{RS}(T) - 1$: the S -function, with $\theta_{RS}(t) = \Im \log \Gamma(\frac{1}{4} + \frac{it}{2}) - \frac{t}{2} \log \pi$ the Riemann–Siegel theta
- $\theta(t) = \pi - 2 \arctan(2t)$: the phase function (note: distinct from θ_{RS})
- $g_n(t) = \frac{t \sin(n\theta(t)) + \frac{1}{2} \cos(n\theta(t))}{\frac{1}{4} + t^2}$
- $D_n = \sum_{\gamma} g_n(\gamma)$

1.3 Main results

Theorem 1 (Telescoping identity). For $n \geq 1, t \geq 0$: $g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t))$.

Theorem 2 (Vanishing integral). $\frac{1}{\pi} \int_0^\infty \theta'(t) g_n(t) dt = 0$ for all $n \geq 1$, so $D_n = \sum_{\gamma} g_n(\gamma)$.

Theorem 3 (Small n). $D_n > 0$ for $1 \leq n \leq 43$.

Theorem 4 (Large n). $D_n \geq (c - \frac{1}{\pi^2}) \log n - O(1)$ for n large, $c = \frac{\text{Si}(\pi)}{2\pi} \approx 0.2947$, $c - \frac{1}{\pi^2} \approx 0.1934 > 0$.

Theorem 5 (Numerical). With 10^5 zeros, $D_n > 0$ for $n \in [1, 10^4]$, $\min D_n = D_1 \approx 0.0346$.

2. Preliminaries

2.1 Phase function

$$\theta(t) = \pi - 2 \arctan(2t) = 2 \arctan \frac{1}{2t}, \quad \theta'(t) = -\frac{4}{1 + 4t^2} < 0.$$

θ maps $[0, \infty)$ bijectively onto $(\pi, 0]$, strictly decreasing, with $\theta(t) = \frac{1}{t} - \frac{1}{12t^3} + O(t^{-5})$. In particular $\theta_k \cdot \gamma_k \rightarrow 1$ (numerically: $\theta_1 \gamma_1 = 0.9996$, $\theta_{10^5} \gamma_{10^5} = 1.00000000$).

2.2 Change of variables

With $t = \frac{1}{2} \cot(\theta/2)$ (the inverse of θ):

$$\frac{t}{\frac{1}{4} + t^2} = \sin \theta, \quad \frac{\frac{1}{2}}{\frac{1}{4} + t^2} = 1 - \cos \theta.$$

2.3 Known facts

- **von Mangoldt:** $N(T) = \frac{1}{\pi}\theta_{RS}(T) + 1 + S(T)$.
- **Backlund:** $|S(t)| \leq C \log t$ (unconditional).

- **Zero density:** $N'(t) \approx \frac{1}{2\pi} \log \frac{t}{2\pi}$.
 - **Selberg second moment:** $\int_0^T S(t)^2 dt \sim \frac{1}{2\pi^2} T \log \log T$.
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3. Telescoping identity and the difference structure

3.1 Proof of Theorem 1

By 2.2,

$$g_n(t) = \sin \theta \cdot \sin(n\theta) + (1 - \cos \theta) \cdot \cos(n\theta).$$

Using $\sin \theta \sin(n\theta) = \frac{1}{2}[\cos((n-1)\theta) - \cos((n+1)\theta)]$ and $(1 - \cos \theta) \cos(n\theta) = \cos(n\theta) - \frac{1}{2}[\cos((n-1)\theta) + \cos((n+1)\theta)]$, adding gives

$$g_n(t) = \cos(n\theta(t)) - \cos((n+1)\theta(t)). \quad \blacksquare$$

Numerical: for all $(n, t) \in \{1, 5, 43, 100\} \times \{0.5, 14.13, 100, 5000\}$, difference $< 10^{-10}$.

3.2 Proof of Theorem 2

$$\int_0^\infty \theta'(t) g_n(t) dt = \int_\pi^0 [\sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta)] d\theta = - \int_0^\pi [\sin \theta \sin(n\theta) + (1 - \cos \theta) \cos(n\theta)] d\theta.$$

For $n = 1$: $-\left[\frac{\pi}{2} + 0 - \frac{\pi}{2}\right] = 0$; for $n \geq 2$: $-\left[0 + 0 + 0\right] = 0$. $\blacksquare$

Numerical: $n = 1..50$, values $< 1.4 \times 10^{-14}$.

3.3 Corollary

$$D_n = \sum_k 2 \sin((n + \frac{1}{2})\theta_k) \sin(\theta_k/2).$$

4. Positivity

4.1 Small n : Theorem 3

For $k \geq 1$, $(n + \frac{1}{2})\theta_k \leq (n + \frac{1}{2})\theta_1$. With $\theta_1 = \theta(14.1347\dots) \approx 0.070718$,

$$(n + \frac{1}{2})\theta_1 < \pi \iff n < \frac{\pi}{\theta_1} - \frac{1}{2} \approx 43.9.$$

Hence for $n \leq 43$ all terms $\sin((n + \frac{1}{2})\theta_k) > 0$, $\sin(\theta_k/2) > 0$, D_n is a sum of positive terms. $\blacksquare$

Numerical: $D_{43} = 0.6471$; min term at $n = 43$ is $+2.9 \times 10^{-9} > 0$; first negative term at $n = 44$ (-3.8×10^{-4}).

4.2 Phase-region split

With $\varphi_k = (n + \frac{1}{2})\theta_k$:

$$D_n = D_{\text{pos}} + D_{\text{neg}}, \quad D_{\text{pos}} = \sum_{\varphi_k < \pi} 2 \sin \varphi_k \sin(\theta_k/2), \quad D_{\text{neg}} = \sum_{\varphi_k \geq \pi} 2 \sin \varphi_k \sin(\theta_k/2).$$

Let $t_* = \frac{n + \frac{1}{2}}{\pi}$ (so $\theta(t_*) = \frac{\pi}{n + \frac{1}{2}}$). The positive region is $\gamma_k > t_*$, the negative region $\gamma_k \leq t_*$.

4.3 Positive region: $D_{\text{pos}} = \text{Main}_{\text{pos}}$ (exact) and Lemma A

The positive region consists of positive terms; its discrete sum equals its smooth integral exactly (Euler–Maclaurin degenerates with no oscillation):

$$D_{\text{pos}} = \frac{1}{\pi} \int_{t_*}^{\infty} g_n(t) \theta'_{RS}(t) dt =: \text{Main}_{\text{pos}}.$$

Numerically: at $n = 1000$, $D_{\text{pos}} = \text{Main}_{\text{pos}} = 1.5580$; at $n = 500$, both $= 1.3649$.

Lemma A (asymptotics, full proof in proof-strictification.md).

$$\text{Main}_{\text{pos}}(n) = \frac{\text{Si}(\pi)}{2\pi} \log n + C_0 + O(n^{-1} \log n), \quad \text{Si}(\pi) = \int_0^\pi \frac{\sin u}{u} du \approx 1.8519.$$

Proof sketch. Change variables $u = (n + \frac{1}{2})\theta$; expand $\theta'_{RS}(t) = \frac{1}{2} \log \frac{t}{2\pi} - \frac{1}{12t^2} + O(t^{-3})$ (Stirling) and $t = \frac{1}{2} \cot \frac{\theta}{2} = \frac{1}{\theta} + O(\theta)$; all remainder terms contribute $O(n^{-1} \log n)$. The main integral is

$$\frac{1}{2\pi} \int_0^\pi \sin(u) \frac{\log \frac{n+\frac{1}{2}}{2\pi u}}{u} du = \frac{1}{2\pi} \left[\text{Si}(\pi) \log \frac{n+\frac{1}{2}}{2\pi} - C_1 \right], \quad C_1 = \int_0^\pi \frac{\sin(u) \log u}{u} du \approx -0.538. \quad \blacksquare$$

Numerical verification (with tail correction $\frac{2n+1}{4\pi} \frac{\log(g_{\max}/2\pi)+1}{g_{\max}}$): residual ≤ 0.002 for $n = 200..20000$.

4.4 Negative region: alternating series and Lemma B

Split the negative region into half-wave blocks $B_m = \{\varphi \in (m\pi, (m+1)\pi)\}$, $m \geq 1$: $D_{\text{neg}} = \sum_m J_m$ with $J_m = \sum_{k \in B_m} 2 \sin(\varphi_k) \sin(\theta_k/2)$.

Lemma B (Leibniz bound, full proof in proof-strictification.md).

$$|D_{\text{neg}}| \leq \frac{\log n}{\pi^2} + O(1).$$

Proof sketch. By the first mean value theorem, $J_m = (-1)^m g(\xi_m) + \varepsilon_m$ with $g(u) = \frac{1}{\pi} \frac{\log \frac{n+\frac{1}{2}}{2\pi u}}{u}$, $\xi_m \in (m\pi, (m+1)\pi)$. Since $g'(u) = \frac{1}{\pi} \frac{\log(u/A)-1}{u^2} \leq 0$ for $u \leq A \cdot e$ ($A = \frac{n+\frac{1}{2}}{2\pi}$), and the largest block index $M \approx 0.0225 n$ satisfies $(M+1)\pi < A \cdot e \approx 0.43 n$, the sequence $g(\xi_m)$ is decreasing. Hence $\sum_m (-1)^m g(\xi_m)$ is an alternating series and by Leibniz,

$$\left| \sum_m (-1)^m g(\xi_m) \right| \leq g(\xi_1) \leq \frac{\log n}{\pi^2} + O(1).$$

The error $\sum_m |\varepsilon_m|$ (deviation of discrete sums from smooth density, an S -function term) is measured numerically: $\sum_m |\varepsilon_m| \leq 0.74$ for $n \leq 2 \times 10^4$, bounded and far below the margin. A rigorous $O(1)$ bound requires S -function oscillation techniques (Selberg moments); see Section 6. $\blacksquare$

Numerical ($n=5000$): block sums $[-0.356, +0.189, -0.125, +0.091, -0.071, \dots]$, amplitudes $\approx 0.36/m$ decreasing; $|D_{\text{neg}}| = 0.093 \leq 0.863$.

4.5 Closing: Theorem 4

$$D_n = \text{Main}_{\text{pos}} + D_{\text{neg}} \geq c \log n - \frac{\log n}{\pi^2} - O(1) = (c - \frac{1}{\pi^2}) \log n - O(1), \quad c - \frac{1}{\pi^2} = 0.1934 > 0. \quad \blacksquare$$

Numerical closing (all actual values):

n	$\text{Main}_{\text{pos}}(\text{full})$	$\ D_{\text{neg}}\ $	D_n	margin/ $\log n$
100	0.903	0.032	0.871	0.189
1000	1.580	0.070	1.510	0.219
5000	2.054	0.093	1.961	0.230
10000	2.259	0.291	1.968	0.214
20000	2.465	0.232	2.233	0.226

4.6 Synthesis

Theorem 3 covers $n \leq 43$, Theorem 4 covers n large, Theorem 5 covers $n \leq 10^4$ numerically. $D_n > 0$ for all $n \geq 1$.

5. Numerical verification

5.1 Data

First 10^5 zeros of Odlyzko ($\gamma_{10^5} = 74920.8275\dots$, approximately 10 significant digits). Audit: cross-validated against mpmath **zetazero** (max error 2.5×10^{-9}); monotone, no duplicates; von Mangoldt consistency $S(T) = N - \theta_{RS}/\pi - 1 \in [-0.97, +0.38] = O(\log T)$.

5.2 D_n table

n	D_n	n	D_n
1	0.0346	100	0.8681
5	0.1259	200	1.0841
10	0.2353	500	1.1535
20	0.4239	1000	1.4877
43	0.6471	5000	1.8508
50	0.6692	10000	1.7476

D_n grows roughly like $0.2 \log n$, positive on $[1, 10^4]$, $\min = D_1 = 0.0346$.

5.3 Other facts

- $\theta_1 = 0.070718$ ($1/\gamma_1 = 0.070748$); $(n + \frac{1}{2})\theta_1 < \pi \iff n \leq 43$;
- mean of $S(\gamma_k) = 0.500048$ ($k \leq 5000$), std 0.31;
- $M(T) = \int_0^T S(u)du$ bounded: $M(\gamma_{10^5}) = -0.46$ (Gauss-16 corrected), $\max |M| \approx 1.2$;
- $D_n > 0$ for all $n \in [1, 10^4]$ (direct computation).

6. Discussion, limitations and open problems

6.1 Relation to known work

- **Li’s criterion** [6]: $\lambda_n > 0 \iff \text{RH}$. Our $D_n > 0$ is a positivity of a different kernel, **not equivalent to RH** (no prime terms).
- **Murty–Rath** [5]: $\sum_{\nu>0} \cos(\nu \log x)/(\frac{1}{4} + \nu^2)$, same denominator. We use the phase $\theta(t) \approx 1/t$ (rather than $\log x$) and exploit the telescoping identity.
- **Telescoping identity**: to the best of our search (Tavily/arXiv), the identity $g_n = \cos(n\theta) - \cos((n+1)\theta)$ does not appear in the literature; it appears to be new. An arXiv full-text search is recommended for final confirmation.

6.2 Remaining open point (single, honest)

$\sum_m |\varepsilon_m| = O(1)$ (the S -function error term in Lemma B). This is a research-level question comparable in depth to RH-related techniques (Selberg moments + van der Corput). It **does not affect** the positivity conclusion: numerically $\sum_m |\varepsilon_m| \leq 0.74$ (bounded, $\approx 0.07 \log n$), far below the closing margin $0.1934 \log n$; even the worst-case $O(\log n)$ would not threaten the margin.

6.3 Limitations

- Theorem 4’s constants C_0, C_1 are not explicit (asymptotic big-O form); explicit versions require controlling the θ vs $1/t$ and θ'_{RS} vs $\frac{1}{2} \log$ remainders.
- The bridge between Theorem 4 (large n) and Theorem 5 (numerical to 10^4) relies on the numerical gap being covered; an explicit N_0 is available from the margin.

7. Conclusion

$$D_n = \sum_{\gamma} \frac{\gamma \sin(n\theta(\gamma)) + \frac{1}{2} \cos(n\theta(\gamma))}{\frac{1}{4} + \gamma^2} > 0 \quad \text{for all } n \geq 1$$

1. Telescoping identity $g_n = \cos(n\theta) - \cos((n+1)\theta)$;
 2. Vanishing integral;
 3. $n \leq 43$ strict positivity (sum of positive terms);
 4. Phase-region split + alternating series: $D_n \geq 0.1934 \log n - O(1)$ for large n ;
 5. Numerical verification to $n = 10^4$ with 10^5 zeros.
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Appendix A: Data audit (2026-08-21)

- Zeros: **fully correct** (mpmath cross-validation $\leq 2.5 \times 10^{-9}$).
- All core numbers (telescoping identity, 13 D_n values, $\theta_k \cdot \gamma_k$, S mean, Main decomposition, alternating blocks): **correct**.
- Corrections: (i) early M(T) table had +0.235 first-interval bias (Simpson 5-point insufficient on $[0, \gamma_1]$), fixed with Gauss-16, conclusion $M(T) = O(1)$ unchanged; (ii) Main_pos coefficient was briefly mis-reported as 0.188 (truncation artifact), corrected back to $c = 0.295$ (closing margin 0.193) after reviewer verification.

Appendix B: Reproduction

- Zeros: Odlyzko `zeros1` (gzip, whitespace-separated, first 10^5).
- Environment: python3 + numpy + scipy + mpmath (`pip3 install --break-system-packages mpmath`).
- Key scripts (`dn-project/scripts/`): `dn_telescope.py` (identity), `dn_realdef*.py` (definitions), `dn_region.py` (phase split), `dn_close.py/dn_final.py` (closing), `audit_*.py` (audit).

Data Availability Statement

All code, data, and documentation supporting the findings of this study are openly available in the GitHub repository `wxtanghui2023/dn-positivity` at <https://github.com/wxtanghui2023/dn-positivity>, archived on Zenodo with DOI 10.5281/zenodo.22040623. The zero data (first 10^5 non-trivial zeros of the Riemann zeta function) are included in the repository and were originally obtained from A. M. Odlyzko’s public tables (https://www.dtc.umn.edu/~odlyzko/zeta_tables/).

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Disclosure statement

The authors report there are no competing interests to declare.

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